How-To Library

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Performance Tuning Guidelines for Microsoft Azure Data Lake Storage Gen2 Connector

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Abstract

You can use Microsoft Azure Data Lake Storage Gen2 Connector to read from or write to Microsoft Azure Data Lake Storage Gen2. This article describes general reference guidelines to help you tune the performance of Microsoft Azure Data lake Storage Gen2 Connector in Cloud Data Integration. It also describes the best practices and case studies that involve the tuning specifications required when you run a mapping to read from or write to Microsoft Azure Data Lake Storage Gen2.

Supported Versions

- Informatica Cloud® Data Integration Microsoft Azure Data Lake Storage Gen2 Connector

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Overview

The performance tuning is an iterative process to analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the key hardware, sizing, and mapping parameters that you can tune to optimize the performance of Microsoft Azure Data Lake Storage Gen2 Connector.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Microsoft Azure Data Lake Storage Gen2 Connector might vary based on individual environments and other parameters even when you use the same data.

Performance sizing parameters

You can tune the hardware parameters based on the data set and use the recommended hardware capacities to optimize the performance of the machine where the Secure Agent runs.

Performance sizing depends on the size of the data set. Consider the following sizing recommendations for the different types of data sets:

Small data set

A data set is considered to be a small data set if you read from and write to less than or equal to 750 MB of data.

For a small data set, you can read 750 MB of data from a flat file source and write to Microsoft Azure Data Lake Storage Gen2 in 00:00:44 seconds. Read data from Microsoft Azure Data Lake Storage Gen2 and write data to a flat file target in 00:01:24 minutes.

The following table describes the sizing recommendations for the system that processes small data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	1
Memory	4 GB
Storage	50 GB

Medium data set

A data set is considered to be a medium data set if you read from and write to less than or equal to 7.5 GB of data.

For a medium data set, you can read 7.5 GB of data from a flat file source and write to Microsoft Azure Data Lake Storage Gen2 in 00:05:57 minutes. Read data from Microsoft Azure Data Lake Storage Gen2 and write data to a flat file target in 00:08:50 minutes.

The following table describes the sizing recommendations for the system that processes medium data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	2
Memory	8 GB
Storage	100 GB

Large data set

A data set is considered to be a large data set if you read from and write to less than or equal to 75 GB of data.

For a large data set, you can read 75 GB of data from a flat file source and write to Microsoft Azure Data Lake Storage Gen2 in 0:55:36 minutes. Read data from Microsoft Azure Data Lake Storage Gen2 and write data to a flat file target in 01:35:21 hours.

The following table describes the sizing recommendations for the system that processes large data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	8

Requirements	Infrastructure
Memory	28 GB
Storage	250 GB

Performance tuning parameters

You can optimize the performance of Microsoft Azure Data Lake Storage Gen2 mappings by tuning the following areas:

- Tune the hardware
- Tune the heap size for the Secure Agent
- Tune the mapping
  - Tune the reader performance
  - Tune the writer performance

Tune the hardware

You can tune the following hardware parameters to optimize the performance of the machine where the Secure Agent runs:

- CPU Frequency
- NIC Card ring buffer size

CPU frequency

Dynamic frequency scaling adjusts the frequency of the processor on-the-fly either for power savings or to reduce heat. Ensure that the CPU operates at least at the base frequency.

When CPUs are under clocked, where they run below the base frequency, the performance degrades by 30% to 40%. Informatica recommends that you work with your IT system administrator to ensure that all the nodes on the cluster are configured to run at their supported base frequency.

To tune the CPU frequency for Intel multi-core processors, perform the following steps:

1. Run the lscpu command to determine the current CPU frequency, base CPU frequency, and the maximum CPU frequency that the processor supports.
2. Request your system administrator to perform the following tasks:
   - Increase the CPU frequency to the supported base frequency.
   - Change the power management setting to **OS Control** at the BIOS level.
3. Run CPU-intensive tests to monitor the CPU frequency in real time and adjust the frequency for improved performance. On Red Hat operating systems, you can install a monitoring tool such as cpupower.
4. Work with your IT department to ensure that the CPU frequency and power management settings are persisted even for future system reboots.

NIC card ring buffer size

NIC configuration is a key factor in network performance tuning. When you deal with large volumes of data, it is crucial that you tune the Receive (RX) and Transmit (TX) ring buffer size. The ring buffers contain descriptors or pointers to the socket kernel buffers that hold the packet data.

You can run the ethtool command to determine the current configuration. For example, run the following command:

```
# ethtool -g eth0
```

The following sections show a sample output:

```
Ring parameters for eth0:

Pre-set maximums:

RX: 2040

RX Mini: 0

RX Jumbo: 8160

TX: 255


Current hardware settings:

RX: 255

RX Mini: 0

RX Jumbo: 0

TX: 255
```

The **Pre-set maximums** section shows the maximum values that you can set for each parameter. The **Current hardware settings** section shows the current configuration details.

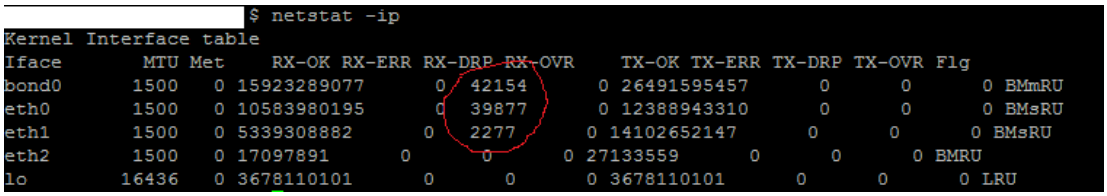
A long buffer size leads to low latency. However, low latency comes at the cost of throughput. For greater throughputs, you must configure large buffer ring sizes for RX and TX.

Informatica recommends that you use the ethtool command to determine the current hardware settings and the maximum supported values. Then, set the values based on the maximum values that are supported for each operating system. For example, if the maximum supported value for RX is 2040, you can use the ethtool command as follows to set the RX value to 2040:

```
# ethtool -G eth0 RX 2040
```

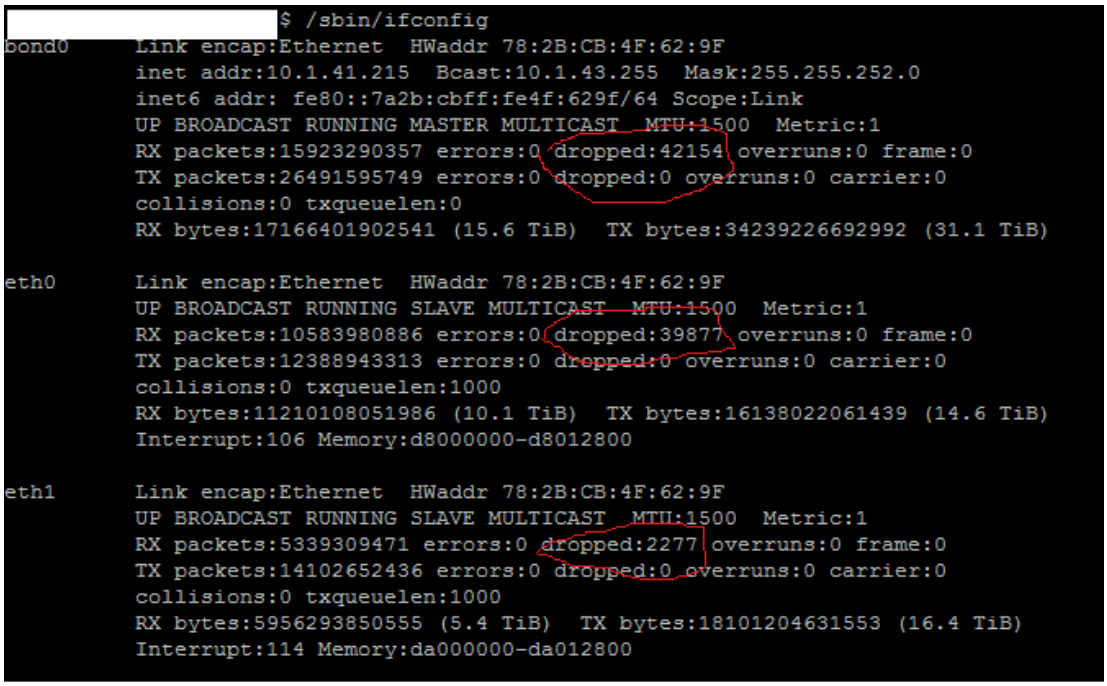
If you set a low ring buffer size for data transfer, packets might get dropped. To find out if packets were dropped, you can use the netstat and ifconfig commands.

The following image shows a sample output of the netstat command:



The RX-DRP column indicates the number of packets that were dropped. Set the RX value such that no packets get dropped and the RX-DRP column shows the values as 0. You might need to test several values to optimize the performance.

The following image shows a sample output of the ifconfig command:



The status messages indicate the number of packets that were dropped.

Tune the heap size for the Secure Agent

To improve the reader and writer performance of medium and large data sets and avoid runtime memory issues, configure the Java heap size for the Secure Agent. The default heap size is 64 MB.

1. Log in to Informatica Intelligent Cloud Services.
2. Select **Administrator** > **Runtime Environments**.
3. On the **Runtime Environments** page, select the Secure Agent for which you want to increase memory from the list of available Secure Agents.
4. On the upper-right corner of the page, click **Edit**.
5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Server.
6. Enter **JVMOption1** as **-Xmx1024m**.
7. Click **Save**.

Best practices

When you tune the reader or writer, you can employ certain best practices to improve the performance.

Best practices for a mapping task

You can employ the following best practices to improve the performance of a mapping task:

- Host the Secure Agent on an Azure Virtual Machine. Azure provides the flexibility to scale or descale based on the workloads and provides the VM resource monitoring.
- For data migration across networks, deploy the Secure Agent close to the source object.

- For data migration within a network, deploy the Secure Agent in the same region or network as the source and target object.
- If no transformations are required to process the data, use mass ingestion tasks to transfer a large number of files.

You can tune the following components to improve the reader and writer performance:

Gzip compression

To move large datasets across the network, you can compress the file to minimize the download time, upload time, and network utilization. File compression uses ~90% of the CPU, as it spawns multiple threads to compress a file. The CPU utilization increases for a very short duration of time, as compression is made 8 to 10 times faster with the parallel gzip implementation.

You can enable the gzip compression in the advanced source and target properties for a mapping task.

Note: The time to download or upload a file depends on the network bandwidth of Secure Agent hosted virtual machine and the distance between the geographical location of the Secure Agent hosted virtual machine and the staging location.

Microsoft Azure Virtual Machine

You can tune the following components of a Microsoft Azure Virtual Machine that hosts the Secure Agent, to improve reader and writer performance:

- **Virtual machine type** - Select the virtual machine type based on the workload and kind of operations to be run on the data set.
  For more information about virtual machine sizes, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/sizes?toc=/azure/virtual-machines/windows/toc.json
- **Disks** - The data is staged in the Secure Agent hosted virtual machine. Ensure that the size of the OS disk is greater than the size of data set. For better performance, use the disks with high IOPS and throughput.
  For more information about types of disks, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/disks-types

  The following table shows the recommended disk type for different data sets:

Parameter	Small data set	Medium data set	Large data set
Disk type	Standard HDD >= S10	Premium SSD >= P4	Premium SSD >= P10

Best practices for a mass ingestion task

You can employ certain best practices to improve the performance of a mass ingestion task for files. The following best practices are for passthrough use cases and do not involve any transformations:

- Host the Secure Agent on an Azure Virtual Machine. Azure provides the flexibility to scale or descale based on the workloads and provides the VM resource monitoring.
- For data migration across networks, deploy the Secure Agent close to the source object.
- For data migration within a network, deploy the Secure Agent in the same region or network as the source and target object.
- Use mass ingestion to read large files from a Microsoft Azure Data Lake Storage Gen2 source and write to a Microsoft Azure Synapse SQL target. Mass ingestion eliminates the time taken to download data to the runtime environment and upload data to the Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 staging location and improves performance significantly.
- Set the batch size same as the number of records in the files when you read data from a Microsoft Azure Data Lake Storage Gen2 source and write data to a Microsoft Azure Synapse SQL target.

- When you move multiple files from on-premises Azure Virtual Machine to Microsoft Azure Data Lake Storage Gen2, ensure that the batch size in the Source Local Folder Object does not exceed the number of VCPUs in the Secure Agent hosted virtual machine.

Gzip compression

To move large datasets across the network, you can compress the file before you upload it to the network to minimize the upload time and network utilization. File compression uses ~90% of the CPU, as it spawns multiple threads to compress a file. The CPU utilization increases for a very short duration of time, as compression is made 8 to 10 times faster with the parallel gzip implementation.

You can enable the gzip compression in the advanced target properties for a mass ingestion task.

The time to upload a file depends on the network bandwidth of Secure Agent hosted virtual machine and the distance between the geographical location of the Secure Agent hosted virtual machine and the staging location.

Scalability test

Use the scalability test to understand the effects of increasing the data volume on the execution time, keeping the mapping level configurations and virtual machine size constant.

Performance is linearly scalable when the increase in data volume is linear to the increase in execution time.

Reader scalability test

The reader scalability test describes the effect of increasing the data volume on the execution time of a mapping when you read data from Microsoft Azure Data Lake Storage Gen2.

The following configurations are considered for the test:

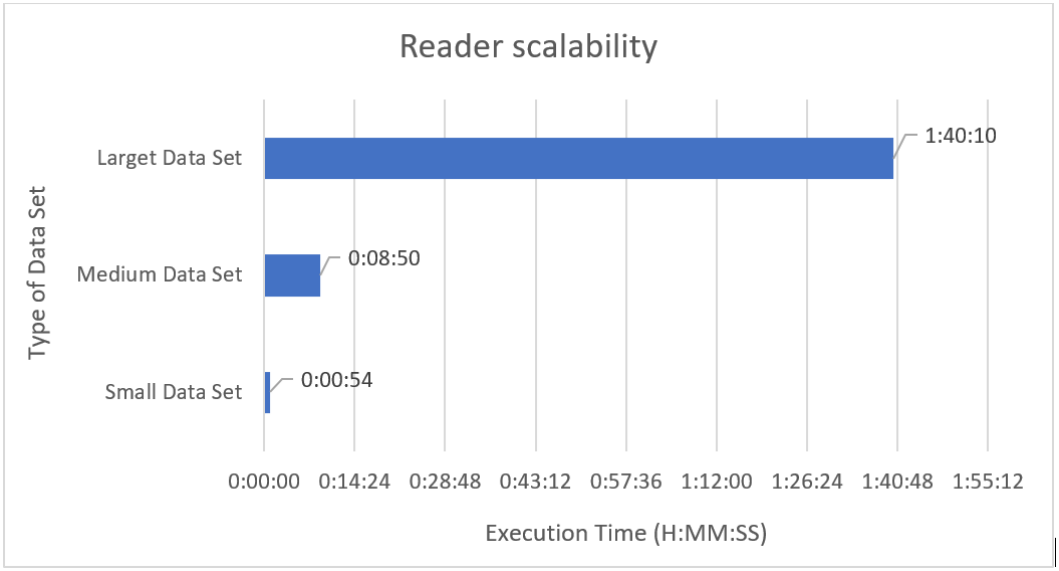
- The source is Microsoft Azure Data Lake Storage Gen2 hosted in the US-EAST-2 region. The target is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region.
- All mappings are passthrough. There are no transformations between the source and target.
- The data set size for small, medium, and large data sets is defined based on the sizing recommendations.
- The Secure Agent is hosted on an Azure Virtual Machine in US-EAST-2 region for all the data sets.

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

Performance graph

The following graph shows the effect of increasing the data volume on the execution time of a reader mapping:



Conclusion

Based on the case study, linear scalability is observed for the reader mapping. When you increase the data volume, the execution time also increases linearly.

Writer scalability test

The writer scalability test describes the effect of increasing the data volume on the execution time of a mapping when you write data to Microsoft Azure Data Lake Storage Gen2.

The following configurations are considered for the test:

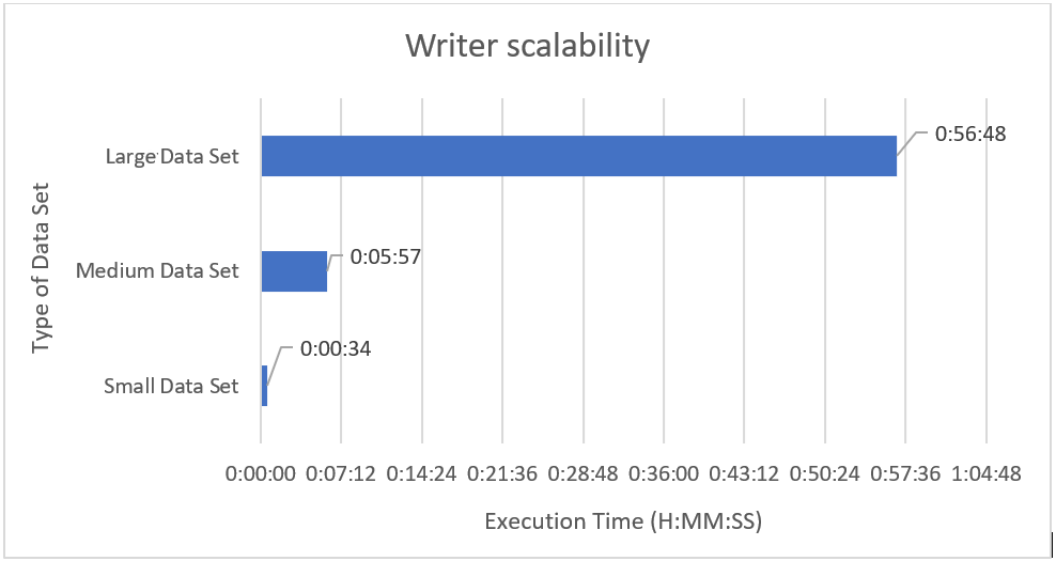
- The source is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region. The target is Microsoft Azure Data Lake Storage Gen2 hosted in the US-EAST-2 region.
- All mappings are passthrough. There are no transformations between the source and target.
- The data set size for small, medium, and large data sets is defined based on the sizing recommendations.
- The Secure Agent is hosted on an Azure Virtual Machine in the US-EAST-2 region for all the data sets.

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

Performance graph

The following graph shows the effect of increasing the data volume on the execution time of writer mapping:



Conclusion

Based on the case study, linear scalability is observed for the writer mapping. When you increase the data volume, the execution time also increases linearly.

Case Study: Mapping task versus mass ingestion task

The case study compares the performance of a mapping task with a mass ingestion task to transfer data.

The following configurations are considered for all the case studies:

- The data set size is 75 GB and 600,000,000 records. The file format is CSV.
- All the mappings are passthrough.
- The Secure Agent is hosted in the same region as the source.

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

Reading data from Microsoft Azure Data Lake Storage Gen2

The case study compares the performance of a mapping task and a mass ingestion task when you read data from Azure Data Lake Storage Gen2.

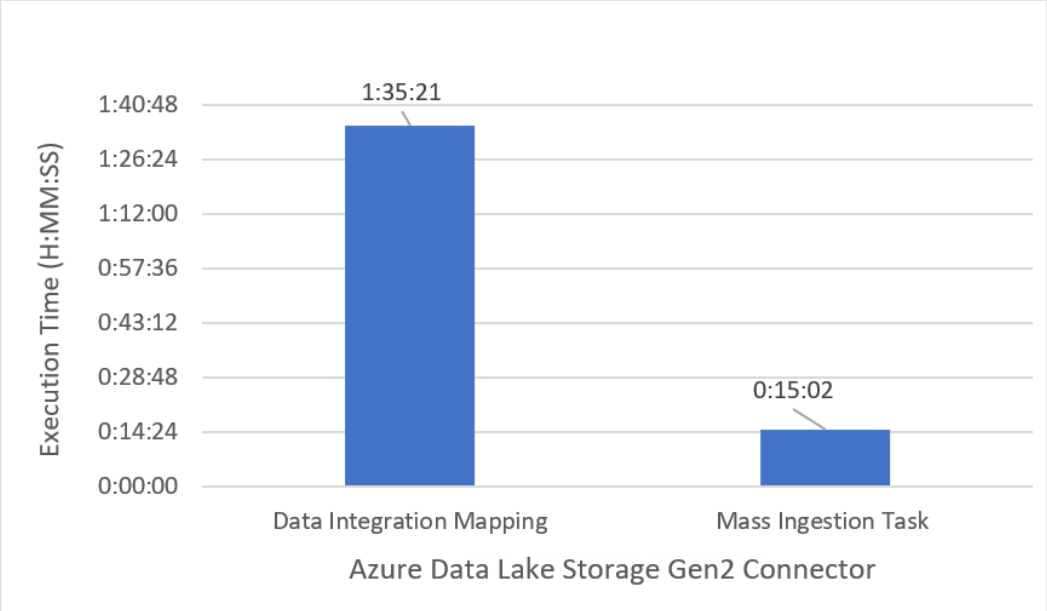
The following configurations are considered for the case study:

- The source is Microsoft Azure Data Lake Storage Gen2 in the US-EAST-2 region. The target is a local folder for mass ingestion task and a flat file data object for mapping task on an Azure Linux Virtual Machine hosted in the US-EAST-2 region.

- The heap size for mass ingestion task is 1024m.
- Compression: None.

Performance graph

The following graph compares the performance of a mapping task and a mass ingestion task when you read data from Azure Data Lake Storage Gen2:



Conclusion

Based on the case study, a significant decrease in the execution time is observed when you use a mass ingestion task to extract data from a Microsoft Azure Data Lake Storage Gen2 source.

The execution time might vary in other environments. The execution time depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

Note: Informatica recommends that you use a mapping task for ETL use cases where you need to transform the extracted data before loading.

Writing data to Microsoft Azure Data Lake Storage Gen2

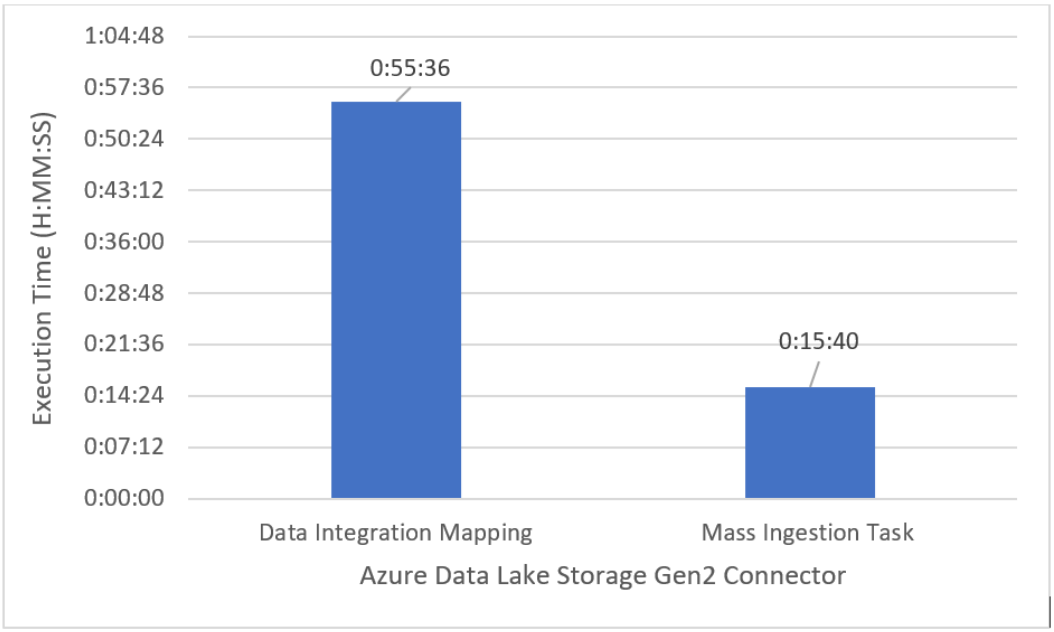
The case study compares the performance of a mapping task and a mass ingestion task when you write data to Microsoft Azure Data Lake Storage Gen2.

The following configurations are considered for the case study:

- The source is a local folder for mass ingestion task and a flat file data object for mapping task on Azure Linux Virtual Machine hosted in the US-EAST-2 region. The target is Microsoft Azure Data Lake Storage Gen2 in the US-EAST-2 region.
- The heap size for mas ingestion task is 1024m.
- Compression: None.

Performance graph

The following graph compares the performance of a mapping task and a mass ingestion task when you write data to Microsoft Azure Data Lake Storage Gen2:



Conclusion

Based on the case study, a significant decrease in the execution time is observed when you use a mass ingestion task to load data to a Microsoft Azure Data Lake Storage Gen2 target.

The execution time might vary in other environments. The execution time depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

Note: Informatica recommends that you use a mapping task for ETL use cases where you need to transform the extracted data before loading.

Author

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